Prime Numbers and the Fundamental Theorem of Arithmetic

- **Definition 1**
  An integer $p > 1$ is a **prime number** if its only positive divisors are 1 and p. An integer $n > 1$ that is not prime is called **composite**.
    - **Example 1.1**
      The first prime numbers are
      $$2, 3, 5, 7, 11, 13, 17, 19, 23, \cdots$$
      The numbers
      $$4, 6, 8, 9, 10, 12, \cdots$$
      are composite.

  Note that 1 is neither prime nor composite.
    - **Proposition 1.2**
      An integer $n > 1$ is composite, if and only if, $n = ab$ for some $a, b \in \mathbb{N}$ with $1 < a < n$ and $1 < b < n$.
        - **Proof 1.2.1**
          ($\Longrightarrow$) Suppose that n is composite, then it has a positive divisor a with $a \neq 1, n$. Hence there exists $b \in \mathbb{N}$ such that $n = ab$ with $1 < a < n$ and $1 < b < n$, because if $b = 1$ then $a = n$.

          ($\Longleftarrow$) Suppose that $n = ab$ for some $a, b \in \mathbb{N}$ with $1 < a < n$ and $1 < b < n$. Then such a factorization makes a a divisor of n with $a \neq 1, n$. Therefore n is composite.
    - **Lemma 1.3**
      Let p be a prime and let $a \in \mathbb{Z}$. If $p \nmid a$, then there exist $x, y \in \mathbb{Z}$ such that $ax + py = 1$.
        - **Proof 1.3.1**
          Consider the set
          $$S = \{ax + py : x, y \in \mathbb{Z} \wedge ax + py > 0\}.$$
          Since $p \nmid a$, we have that $a \neq 0$. Therefore $|a| \in S$ because $|a| = a \cdot 1 + p \cdot 0 > 0$ if $a > 0$, or $|a| = a \cdot (-1) + p \cdot 0 > 0$ if $a < 0$.
          Now, as $S \subseteq \mathbb{N}$ and $S \neq \varnothing$, by the well-ordering axiom S has a first element d. Therefore $d = ax_0 + py_0 > 0$. If we divide a by d, then by the division algorithm we have that
          $$a = dq + r \text{ with } 0 \leqslant r < |d| = d.$$
          Hence
          $$r = a - dq = a - (ax_0 + py_0)\, q = a\,(1 - x_0 q) + p\,(-y_0 q).$$
          If $r > 0$, then $r \in S$, which contradicts the minimality of d. Therefore we obtain that $r = 0$ and $a = dq$, that is, $d \mid a$. Using the same argument, dividing p by d shows that $d \mid p$. Now, as p is prime we have that $d = 1 \vee d = p$, but as $p \nmid a$ we have that $d = 1$. Therefore we obtain that $ax_0 + py_0 = 1$.
    - **Corollary 1.4**
      (1) Let p be a prime and let $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a \vee p \mid b$.
        - **Proof 1.4.1**
          Suppose that $p \nmid a$, then by Lemma 1.3 there exist $x, y \in \mathbb{Z}$ such that $ax + py = 1$, hence
          $$abx + pby = b.$$
          Since $p \mid ab$, there exists $k \in \mathbb{Z}$ such that $ab = kp$, so
          $$b = abx + pby = kpx + pby = (kx + by)p \text{ with } kx + by \in \mathbb{Z}.$$
          Therefore $p \mid b$.

(2) Let p be a prime and $a_1, a_2, \cdots, a_n \in \mathbb{Z}$. If $p \mid (a_1 a_2 \cdots a_n)$, then $p \mid a_i$ for some $1 \leqslant i \leqslant n$. In particular, if all the a_j are prime, then $p = a_i$ for some $1 \leqslant i \leqslant n$.

- **Proof 1.4.2**
  By induction.
  1. Base step
     If $n = 1$ then the result is true.
  2. Inductive step
     Suppose that the result is true for n.

$$\begin{aligned} p \mid (a_1 a_2 \cdots a_n a_{n+1}) &\implies p \mid (a_1 a_2 \cdots a_n)\, a_{n+1} \\ &\implies p \mid (a_1 a_2 \cdots a_n) \vee p \mid a_{n+1} \\ &\implies p \mid a_i \text{ for some } 1 \leqslant i \leqslant n \vee p \mid a_{n+1} \\ &\implies p \mid a_j \text{ for some } 1 \leqslant j \leqslant n+1. \end{aligned}$$

     Therefore the result is true for $n + 1$.
     If all the a_j are prime and $p \mid a_i$, then by definition of a prime we have that $p = a_i$.

- **Theorem 2 (Fundamental theorem of arithmetic)**Every integer $n \geqslant 2$ can be written in a unique form up to rearrangements as a product

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$$

where $p_1, p_2, \cdots, p_r$ are distinct primes and $\alpha_1, \alpha_2, \cdots, \alpha_r \in \mathbb{N}$. For each i, the notation $\nu_{p_i}(n) = \alpha_i$ denotes the exponent of the prime p_i in the factorization of n.

  - **Proof 2.1**
    1. Existence
       Consider the set

$$A = \{m \in \mathbb{N} : m \geqslant 2 \wedge m \text{ cannot be written as a product of primes}\}.$$

       If $A \neq \varnothing$, then by the well-ordering axiom A has a first element n. Then n is not a prime. As n is composite, there exist $a, b \in \mathbb{N}$ such that $n = ab$ with $1 < a < n$ and $1 < b < n$. As n is the first element of A, we have that a and b can be written as a product of primes, and therefore n can be written as a product of primes, which is a contradiction because $n \in A$. Therefore $A = \varnothing$ and thus every integer $\geqslant 2$ can be written as a product of prime numbers.
    2. Uniqueness
       Suppose that the result is false, that is, there exists an integer $\geqslant 2$ that can be written in two different forms as a product of primes. Then the set

$$B = \{m \in \mathbb{N} : m \geqslant 2 \wedge m \text{ can be written in two different forms as a product of primes}\}$$

       is nonempty, then by the well-ordering axiom B has a first element n, then

$$p_1 p_2 \cdots p_r = n = q_1 q_2 \cdots q_s.$$

       Since $p_1 \mid q_1 q_2 \cdots q_s$, then $p_1 \mid q_i$ for some $1 \leqslant i \leqslant s$ by $\text{Corollary } 1.4(2)$, and by commutativity of the product we may assume that $i = 1$ and so $p_1 = q_1$. Hence by the cancellation law we have

$$p_2 p_3 \cdots p_r = q_2 q_3 \cdots q_s < n.$$

       Therefore $p_2 p_3 \cdots p_r$ and $q_2 q_3 \cdots q_s$ are two different factorizations of a number $< n$, which contradicts the minimality of n. Therefore $B = \varnothing$, that is, every integer $\geqslant 2$ can be written in a unique form as a product of primes.
  - **Example 2.2**
    Find the prime factorization of 224 and 1260.

$$224 = 2^5 \cdot 7 \quad 1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7.$$

- **Corollary 2.3**
  Every integer $n \leqslant -2$ can be written in a unique form up to rearrangements as a product

$$n = (-1)p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$$

  where $p_1, p_2, \cdots, p_r$ are distinct primes and $\alpha_1, \alpha_2, \cdots, \alpha_r \in \mathbb{N}$.
  - **Proof 2.3.1**
    Since $n \leqslant -2$ then $-n \geqslant 2$, and by the fundamental theorem of arithmetic

$$-n = p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$$

    where $p_1, p_2, \cdots, p_r$ are distinct primes and $\alpha_1, \alpha_2, \cdots, \alpha_r \in \mathbb{N}$.
    Hence

$$n = (-1)p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}.$$

- **Proposition 2.4**
  Suppose $x, y \in \mathbb{N}$, then $\nu_p(xy) = \nu_p(x) + \nu_p(y)$ for any prime p.
  - **Proof 2.4.1**
    Consider the prime factorizations of x and y. Let

$$\begin{aligned} x &= p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r} \quad \alpha_k \in \mathbb{N}_0, \\ y &= p_1^{\beta_1}p_2^{\beta_2}\cdots p_r^{\beta_r} \quad \beta_k \in \mathbb{N}_0. \end{aligned}$$

    Then we have

$$xy = p_1^{\alpha_1+\beta_1}p_2^{\alpha_2+\beta_2}\cdots p_r^{\alpha_r+\beta_r}.$$

    By the uniqueness of the prime factorization of xy, for each $p = p_k$, we have that

$$\nu_p(xy) = \alpha_k + \beta_k = \nu_p(x) + \nu_p(y).$$

- **Theorem 3 (Euclid's theorem)**
  The set of prime numbers is infinite.
  - **Proof 3.1**
    Suppose that there are only finitely many primes $p_1, p_2, \cdots, p_n$ and consider

$$N = p_1p_2\cdots p_n + 1.$$

    Since $N > 1$, by the fundamental theorem of arithmetic we have a prime divisor q of N. Then $q = p_i$ for some $1 \leqslant i \leqslant n$, hence q divides both N and $p_1p_2\cdots p_n$, and therefore q divides $N - p_1p_2\cdots p_n = 1$, which is impossible because q is a prime. Therefore there are infinitely many primes.